Compliance sheet — star-doppler

Everything a supplier qualification, a legal review or an export check usually asks about star-doppler 0.1.0, in one document.

1. Identity

Document	STAR-PKG-STAR-DOPPLER-EN · v1.3 · 2026-10-16
Package	star-doppler
Version	0.1.0
Capability	star.link.doppler
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_doppler

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import
hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_doppler-
0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_doppler-0.1.0-py3-none-any.whl	a55b3f09baaad2635faa4e06c2e0f7e5b06c7e5ca0632eb80d559ea2af4d0c3d
star_doppler.py	49a6510401642b89f441d9ad486cc43a44af31efb7d038f876fdf889141e57fa
star_doppler-0.1.0-source.zip	5c579dde30a0154cb3f611e305be65fc5c963a0f03a4969cc3525d2a97a380e2

5. What it does, and its limits

Range and range rate between two states and the frequency received over that link, by the longitudinal relativistic formula or the first-order one; on 600 pairs of states and range rates up to 0.9 c the results agree with NAIF SPICE (vnorm, dvnorm), astropy's Doppler equivalencies, NumPy and mpmath at 40 digits within 8.3e-16 of their natural scale.

A geometric, instantaneous, one-way result: no light-time correction (the states must already refer to the right instants), no two-way or coherent turn-around ratio, no transverse relativistic term (the relativistic formula is exact only for motion along the line of sight; the neglected term is of order $(v/c)^2$, the same order as the difference between the two formulas), no gravitational shift, no troposphere or ionosphere, no oscillator error. For a low orbit those omissions are of the order of a hertz at S band: enough to plan a receiver bandwidth, not to do orbit determination.

- `range_and_rate(r_observer, v_observer, r_target, v_target)` returns the distance and the component of the relative velocity along the line of sight, positive when the two move apart. Any frame and units, the same for all.
- `received_frequency(frequency, range_rate, relativistic=True)` returns f sqrt((1 − b)/(1 + b)) with b = range_rate / c, or the first-order f (1 − b) with `relativistic=False`; `C_KM_S` = 299792.458 km/s.
- Measured on 600 pairs of states and range rates up to 0.9 c: range, range rate and both frequencies agree with NAIF SPICE (`vnorm`, `dvnorm`), astropy's Doppler equivalencies, NumPy and mpmath at 40 digits within 8.3e-16 of their natural scale (the relative speed for the range rate).

- Every function returns floats or raises `ValueError`: vectors that are not three finite numbers within 1e15, coincident positions, a frequency that is not positive and finite, a range rate that reaches the speed of light, `relativistic` that is not a bool.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_doppler_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-doppler}: Range, range rate and one-way Doppler shift},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_doppler/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

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